

THE h-th MOMENT OF THE TRACE OF A NONCENTRAL WISHART MATRIX

D.G. Nel*

University of the Orange Free State

1. Introduction

The purpose of this paper is to derive the moments of $\text{tr } A$ when A ($p \times p$) has a non-central Wishart distribution with n degrees of freedom and noncentrality parameter Ω . The density of A is given by: (Constantine 1963)

$$(1.1) \quad (1/\Gamma_p(\frac{1}{2}n)) |2\Sigma|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\Sigma^{-1}A) {}_0F_1(\frac{1}{2}n; \frac{1}{4}\Omega\Sigma^{-1}A)$$

where $\text{etr } D = \exp(\text{tr } D)$ and ${}_0F_1$ is a hypergeometric function of matrix argument. (Herz 1955).

The zonal polynomial $C_K(A)$ can be written as: (James (1964)

$$C_K(A) = x_{[2k]}(1) 2^k k! \cdot Z_K(A)/(2k)!$$

where $Z_K(A)$ has been tabulated (James 1964) in terms of the esf. (elementary symmetric functions) of the roots of A for every partition $K = (k_1, \dots, k_p)$ of $k = 1, 2, \dots, 6$.

If we denote the k -th esf. of the roots of A by $\text{tr}_k A$ and use the fact that:

$$Z_{1^k}(A) = k! \cdot \text{tr}_k A \text{ we get:}$$

$$(1.2) \quad \text{tr } A = Z_1(A) = C_1(A) \text{ when the partition } K = 1^k \text{ is considered for } k = 1.$$

The h -th moment of $\text{tr } A$ can be written as:

$$(1.3) \quad E(\text{tr}^h A) = (1/\Gamma_p(\frac{1}{2}n)) |2\Sigma|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K (1/(\frac{n}{2})_K k!) |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\Sigma^{-1}A) C_K(\frac{1}{4}\Omega\Sigma^{-1}A) C_1^h(A) dA.$$

We shall find the solution to this integral only for the special case $\Sigma = \sigma^2 I_p$.

To simplify the multiplication of the zonal polynomials in the integrand we use the fact that:

$$(1.4) \quad [C_1(A)]^k = \sum_K C_K(A)$$

which follows from the definition of zonal polynomials and (1.2) for $h = 1$.

* This research was supported financially by the Central Research Fund of the University of the O.F.S.

2. The calculation of the moments of $\text{tr } A$ when $A \sim W'(\sigma^2 I, n, \Sigma)$

From equation (1.3) it follows for $\Sigma = \sigma^2 I$:

$$(2.1) \quad E(\text{tr}^h A) = (1/\Gamma_p(\frac{1}{2}n)) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K (1/(\frac{n}{2})_K k!) \\ \cdot \int_{A \geq 0} |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\sigma^{-2}A) C_1^h(A) C_K(\frac{1}{4}\sigma^{-2}\Omega) dA.$$

Since the integral is clearly symmetric in Ω we are allowed to transform:

$$\Omega \rightarrow H' \Omega H \text{ where } H \in O(p)$$

But

$$\int_{O(p)} C_K(H' \Omega H) dH = C_K(\Omega) C_K(A)/C_K(I_p) \quad (\text{James 1964})$$

thus (2.1) can be written as:

$$(2.2) \quad E(\text{tr}^h A) = (1/\Gamma_p(\frac{1}{2}n)) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega)/(C_K(I_p)(\frac{n}{2})_K k!) \\ \cdot \int_{A \geq 0} |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\sigma^{-2}A) C_1^h(A) C_K(\frac{1}{2}\sigma^{-2}A) dA.$$

Using (1.4) we get:

$$E(\text{tr}^h A) = 1/\Gamma_p(\frac{1}{2}n) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega)/(C_K(I_p)(\frac{n}{2})_K 2^k \sigma^2 k!) \\ \cdot \int_{A \geq 0} |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\sigma^{-2}A) \sum_H C_H(A) C_K(A) dA.$$

$$(2.3) \quad E(\text{tr}^h A) = \\ (1/\Gamma_p(\frac{1}{2}n)) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega)/(C_K(I_p)(\frac{n}{2})_K 2^k \sigma^2 k!) \\ \cdot \int_{A \geq 0} |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\sigma^{-2}A) \sum_H \sum_{Q \in H} C_Q(A) dA.$$

Where Q is a partition of $q = h+k$ and $C_H(A) C_K(A) = \sum_Q g_{H,K}^Q C_Q(A)$

$$(2.4) \quad E(\text{tr}^h A) = \\ (1/\Gamma_p(\frac{1}{2}n)) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega)/(C_K(I_p)(\frac{n}{2})_K 2^k \sigma^2 k!) \\ \cdot \sum_H \sum_Q g_{H,K}^Q \Gamma_p(\frac{1}{2}n)(\frac{n}{2})_Q C_Q(2\sigma^2 I_p) |1/2\sigma^2 I_p|^{-\frac{1}{2}n}, (\text{Constantine 1963}) \\ = \sigma^2 h \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} a'_k \sum_K a''_K \sum_H \sum_Q g_{H,K}^Q a'''_Q$$

where $a'_k = (1/2^{2k} k!) , a''_K = Z_K(\frac{1}{2}\Omega) / ((\frac{n}{2})_K (\frac{p}{2})_K)$

and $a'''_Q = g_{H,K} (\frac{n}{2})_Q (\frac{p}{2})_Q \times [2Q]^{(1)} 2^{3Q} q! / ((2q)!))$

Let us consider the special case $h = 2$:

To calculate (2.4) we use the following table:

k	K	H	Q	
0	0			$a'_0 = 1 , a''_0 = 1$
		2	2	$a'''_2 = M_0 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{p}{2}) (\frac{p}{2} + 1) , M_0 = \frac{2^6 2!}{4!}$
		1^2	1^2	$a'''_{1^2} = 2 M_0 (\frac{n}{2}) (\frac{n}{2} - \frac{1}{2}) (\frac{p}{2}) (\frac{p}{2} - \frac{1}{2})$
1	1			$a'_1 = \frac{1}{4} , a''_1 = \text{tr}_1 (\frac{1}{2}\Omega) / ((\frac{n}{2}) (\frac{p}{2}))$
		2	3	$a'''_Q = M_1 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} + 2) (\frac{p}{2}) (\frac{p}{2} + 1) \cdot (\frac{p}{2} + 2) , M_1 = \frac{2^9 \cdot 3!}{6!}$
			2,1	$= 4M_1 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} - \frac{1}{2}) (\frac{p}{2}) (\frac{p}{2} + 1) (\frac{p}{2} - \frac{1}{2})$
		1^2	2,1	$= 5M_1 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} - \frac{1}{2}) (\frac{p}{2}) (\frac{p}{2} + 1) (\frac{p}{2} - \frac{1}{2})$
			1^3	$= 5M_1 (\frac{n}{2}) (\frac{n}{2} - \frac{1}{2}) (\frac{n}{2} - 1) (\frac{p}{2}) (\frac{p}{2} - \frac{1}{2}) (\frac{p}{2} - 1)$
2				$a'_2 = 1/32 , M_2 = \frac{2^{12} \cdot 4!}{9!}$
	2			$a''_2 = 3\text{tr}_1^2 (\frac{1}{2}\Omega) - 4\text{tr}_2 (\frac{1}{2}\Omega) / ((\frac{n}{2}) (\frac{n}{2} + 1) (\frac{p}{2}) (\frac{p}{2} + 1))$
		2	4	$a'''_Q = 9M_2 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} + 2) (\frac{n}{2} + 3) (\frac{p}{2}) (\frac{p}{2} + 1) (\frac{p}{2} + 2) \cdot (\frac{p}{2} + 3)$
			3,1	$= 40M_2 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} + 2) (\frac{n}{2} - \frac{1}{2}) (\frac{p}{2}) (\frac{p}{2} + 1) (\frac{p}{2} + 2) \cdot (\frac{p}{2} - \frac{1}{2})$
			2^2	$= 56M_2 (\frac{n}{2}) (\frac{n}{2} + 1) (\frac{n}{2} - \frac{1}{2}) (\frac{n}{2} + \frac{1}{2}) (\frac{p}{2}) (\frac{p}{2} + 1) (\frac{p}{2} - \frac{1}{2}) \cdot (\frac{p}{2} + \frac{1}{2})$

k	K	H	Q	
2	2	1 ²	3,1	$= 70M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \cdot$ $\left(\frac{p}{2} - \frac{1}{2}\right)$
			2,1 ²	$= 140M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} - \frac{1}{2}\right) \cdot$ $\left(\frac{p}{2} - 1\right)$
	1 ²		a'' ₁ ²	$= 2\text{tr}_2 \left(\frac{1}{2}\Omega\right) / \left(\left(\frac{n}{2}\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} - \frac{1}{2}\right)\right)$
		2	3,1	$a'''_Q = 70M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \cdot$ $\left(\frac{p}{2} - \frac{1}{2}\right)$
			2,1 ²	$= 140M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} - \frac{1}{2}\right) \cdot$ $\left(\frac{p}{2} - 1\right)$
		1 ²	2 ²	$= 70M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} - \frac{1}{2}\right) \cdot$ $\left(\frac{p}{2} + \frac{1}{2}\right)$
		2	2,1 ²	$= 224M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} - \frac{1}{2}\right) \cdot$ $\left(\frac{p}{2} - 1\right)$
			1 ⁴	$= 126M_2 \left(\frac{n}{2}\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{n}{2} - 3\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} - 1\right) \cdot$ $\left(\frac{p}{2} - \frac{3}{2}\right)$
3				$a'_3 = 1/384, M_3 = 2^{15} 5! / (10! \cdot 10)$
	3			$a''_3 = 15\text{tr}_1^3 \left(\frac{1}{2}\Omega\right) - 36 \text{tr}_1 \left(\frac{1}{2}\Omega\right) \cdot \text{tr}_2 \left(\frac{1}{2}\Omega\right) + 24\text{tr}_3 \left(\frac{1}{2}\Omega\right) /$ $\left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right)$
		2	5	$a''''_Q = 10M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} + 4\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \cdot$ $\left(\frac{p}{2} + 3\right) \left(\frac{p}{2} + 4\right)$

k	K	H	Q	
3	3	2	4,1	$= 56M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right),$ $\left(\frac{p}{2} + 3\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
			3,2	$= 144M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right),$ $\left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + \frac{1}{2}\right)$
		1 ²	4,1	$a_Q''' = 105M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right),$ $\left(\frac{p}{2} + 3\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
			3,1 ²	$= 315M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right),$ $\left(\frac{p}{2} + 2\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} - 1\right)$
	2,1			$a_{2,1}'' = 4\text{tr}_1 \left(\frac{1}{2}\Omega\right) \text{tr}_2 \left(\frac{1}{2}\Omega\right) - 6\text{tr}_3 \left(\frac{1}{2}\Omega\right) / \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right),$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
		2	4,1	$a_Q''' = 189M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right),$ $\left(\frac{p}{2} + 3\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
			3,2	$= 336M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right),$ $\left(\frac{p}{2} + 2\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + \frac{1}{2}\right)$
			3,1 ²	$= 525M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right),$ $\left(\frac{p}{2} + 2\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} - 1\right)$
		2 ² ,1		$= 840M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right),$ $\left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + \frac{1}{2}\right) \left(\frac{p}{2} - 1\right)$
		1 ²	3,2	$= 420M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2} + 1\right),$ $\left(\frac{p}{2} + 2\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + \frac{1}{2}\right)$

k	K	H	Q	
3	2,1	1 ²	3,1 ²	$=960M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}+2\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}+2\right) \left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right)$
			2 ² ,1	$=1050M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}+\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}+\frac{1}{2}\right) \left(\frac{p}{2}-1\right)$
			2,1 ³	$=1350M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{n}{2}-\frac{3}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right) \left(\frac{p}{2}-\frac{3}{2}\right)$
	1 ³			$a_{1^3}'' = 6\text{tr}_3 \left(\frac{1}{2}\Omega\right) / \left(\frac{n}{2}\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right)$
		2	3,1 ²	$a_Q''' = 450M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}+2\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}+2\right) \left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right)$
		1 ²	2,1 ³	$=600M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{n}{2}-\frac{3}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right) \left(\frac{p}{2}-\frac{3}{2}\right)$
			2 ² ,1	$=630M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}+\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}+\frac{1}{2}\right) \left(\frac{p}{2}-1\right)$
			2,1 ³	$=1050M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{n}{2}-\frac{3}{2}\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right),$ $\left(\frac{p}{2}-\frac{1}{2}\right) \left(\frac{p}{2}-1\right) \left(\frac{p}{2}-\frac{3}{2}\right)$
				$=420M_3 \left(\frac{n}{2}\right) \left(\frac{n}{2}-\frac{1}{2}\right) \left(\frac{n}{2}-1\right) \left(\frac{n}{2}-\frac{3}{2}\right) \left(\frac{n}{2}-2\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}-\frac{1}{2}\right),$ $\left(\frac{p}{2}-1\right) \left(\frac{p}{2}-\frac{3}{2}\right) \left(\frac{p}{2}-2\right)$
4				$a_4' = 1/6144 \quad , \quad M_4 = 2^{18} 6! / (35 \cdot 12!)$
	4			$a_4'' = 105\text{tr}_1^4 \left(\frac{1}{2}\Omega\right) - 360\text{tr}_1^2 \left(\frac{1}{2}\Omega\right) \text{tr}_2 \left(\frac{1}{2}\Omega\right) + 144\text{tr}_2^2 \left(\frac{1}{2}\Omega\right) +$ $288 \text{tr}_1 \left(\frac{1}{2}\Omega\right) \text{tr}_3 \left(\frac{1}{2}\Omega\right) - 192\text{tr}_4 \left(\frac{1}{2}\Omega\right) /$ $\left(\frac{n}{2}\right) \left(\frac{n}{2}+1\right) \left(\frac{n}{2}+2\right) \left(\frac{n}{2}+3\right) \left(\frac{p}{2}\right) \left(\frac{p}{2}+1\right) \left(\frac{p}{2}+2\right) \left(\frac{p}{2}+3\right)$

k	K	H	Q	
4	4	2	6	$a'''_Q = 35M_4 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} + 4\right) \left(\frac{n}{2} + 5\right) \left(\frac{p}{2}\right) \cdot$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \left(\frac{p}{2} + 3\right) \left(\frac{p}{2} + 4\right) \left(\frac{p}{2} + 5\right)$
			5,1	$= 240M_4 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} + 4\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \cdot$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \left(\frac{p}{2} + 3\right) \left(\frac{p}{2} + 4\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
			4,2	$= 880M_4 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} + \frac{1}{2}\right) \left(\frac{p}{2}\right) \cdot$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \left(\frac{p}{2} + 3\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} + \frac{1}{2}\right)$
		1 ²	5,1	$= 462M_4 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} + 4\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{p}{2}\right) \cdot$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \left(\frac{p}{2} + 3\right) \left(\frac{p}{2} + 4\right) \left(\frac{p}{2} - \frac{1}{2}\right)$
			4,1 ²	$= 1848M_4 \left(\frac{n}{2}\right) \left(\frac{n}{2} + 1\right) \left(\frac{n}{2} + 2\right) \left(\frac{n}{2} + 3\right) \left(\frac{n}{2} - \frac{1}{2}\right) \left(\frac{n}{2} - 1\right) \left(\frac{p}{2}\right) \cdot$ $\left(\frac{p}{2} + 1\right) \left(\frac{p}{2} + 2\right) \left(\frac{p}{2} + 3\right) \left(\frac{p}{2} - \frac{1}{2}\right) \left(\frac{p}{2} - 1\right)$

Thus when $h = 2$, eqn. (2.4) becomes:

$$\text{for } k = 0 : \sigma^4 \text{etr}(-\frac{1}{2}\Omega) M_0 \cdot (1/2^4) (3n^2 p^2 + 6np)$$

$$= \sigma^4 \text{etr}(-\frac{1}{2}\Omega) np(np + 2)$$

$$\text{for } k = 1 : \sigma^4 \text{etr}(-\frac{1}{2}\Omega) \left(\frac{1}{4}\right) \text{tr}_1 \left(\frac{1}{2}\Omega\right) M_1 (1/2^4) (15n^2 p^2 + 90np + 120)$$

$$= \sigma^4 \text{etr}(-\frac{1}{2}\Omega) \text{tr}_1 \left(\frac{1}{2}\Omega\right) (np + 2) (np + 4)$$

$$\text{for } k = 2 : \sigma^4 \text{etr}(-\frac{1}{2}\Omega) \cdot (1/32) \cdot M_2 \cdot (1/2^4) [(3\text{tr}_1^2(\frac{1}{2}\Omega) - 4\text{tr}_2(\frac{1}{2}\Omega)) (315n^2 p^2$$

$$+ 3150np + 7560) + 2\text{tr}_2 \left(\frac{1}{2}\right) (820n^2 p^2 + 6300np + 15120)]$$

$$= \sigma^4 \text{etr}(-\frac{1}{2}\Omega) \left(\frac{1}{2}\right) \text{tr}_1^2 \left(\frac{1}{2}\Omega\right) (np + 4) (np + 6)$$

$$\text{for } k = 3 : \sigma^4 \text{etr}(-\frac{1}{2}\Omega) \cdot (1/384) M_3 \cdot (1/2^4) \cdot$$

$$[(15\text{tr}_1^3(\frac{1}{2}\Omega) - 36\text{tr}_1 \left(\frac{1}{2}\Omega\right) \text{tr}_2 \left(\frac{1}{2}\Omega\right) + 24\text{tr}_3 \left(\frac{1}{2}\Omega\right)) \cdot$$

$$\begin{aligned}
& (630n^2p^2 + 8820np + 30240) \\
& + (4\text{tr}_1(\tfrac{1}{2}\Omega) \text{tr}_2(\tfrac{1}{2}\Omega) - 6\text{tr}_3(\tfrac{1}{2}\Omega)) (5670n^2p^2 + 79380np + 272160) \\
& + (6\text{tr}_3(\tfrac{1}{2}\Omega) (3150n^2p^2 + 44100np + 151200)) \\
& = \sigma^4 \text{etr}(-\tfrac{1}{2}\Omega) (1/6) \text{tr}_1^3(\tfrac{1}{2}\Omega) (np + 6) (np + 8)
\end{aligned}$$

If we assume that the terms containing $\text{tr}_1^2(\tfrac{1}{2}\Omega)$, $\text{tr}_2(\tfrac{1}{2}\Omega)$, $\text{tr}_2^2(\tfrac{1}{2}\Omega)$, $\text{tr}_1(\tfrac{1}{2}\Omega)$

$\text{tr}_3(\tfrac{1}{2}\Omega)$ and $\text{tr}_4(\tfrac{1}{2}\Omega)$ will also be cancelled, we have from the above table

for $k = 4$: $\sigma^4 \text{etr}(-\tfrac{1}{2}\Omega) (1/24) \text{tr}_1^4(\tfrac{1}{2}\Omega) (np + 8) (np + 10)$

Thus when $h = 2$ we have for $k = 0, 1, 2, 3, 4, \dots$

$$\begin{aligned}
(2.5) \quad E(\text{tr}^2 A) &= \sigma^4 \text{etr}(-\tfrac{1}{2}\Omega) [np(np + 2) + \text{tr}_1(\tfrac{1}{2}\Omega) (np + 2) (np + 4) \\
&+ 1/2! \text{tr}_1^2(\tfrac{1}{2}\Omega) \cdot (np + 4)(np + 6) + 1/3! \text{tr}_1^3(\tfrac{1}{2}\Omega)(np + 6)(np + 8) + 1/4! \text{tr}_1^4(\tfrac{1}{2}\Omega) \\
&\cdot (np + 8)(np + 10) + \dots]
\end{aligned}$$

By constructing similar tables as above it was also possible to prove for $h = 1, 3$ and 4 that:

$$\begin{aligned}
(2.6) \quad E(\text{tr} A) &= \sigma^2 \text{etr}(-\tfrac{1}{2}\Omega) [np + \text{tr}_1(\tfrac{1}{2}\Omega)(np + 2) + 1/2! \text{tr}_1^2(\tfrac{1}{2}\Omega)(np + 4) \\
&+ 1/3! \text{tr}_1^3(\tfrac{1}{2}\Omega)(np + 6) + 1/4! \text{tr}_1^4(\tfrac{1}{2}\Omega)(np + 8) + \dots]
\end{aligned}$$

and

$$\begin{aligned}
(2.7) \quad E(\text{tr}^3 A) &= \sigma^6 \text{etr}(-\tfrac{1}{2}\Omega) [np(np + 2)(np + 4) + \text{tr}_1(\tfrac{1}{2}\Omega)(np + 2)(np + 4)(np + 6) \\
&+ 1/2! \text{tr}_1^2(\tfrac{1}{2}\Omega)(np + 4)(np + 6)(np + 8) + 1/3! \text{tr}_1^3(\tfrac{1}{2}\Omega) \\
&\cdot (np + 6)(np + 8)(np + 10) \dots]
\end{aligned}$$

and

$$\begin{aligned}
(2.8) \quad (\text{tr}^4 A) &= \sigma^8 \text{etr}(-\tfrac{1}{2}\Omega) [np(np + 2)(np + 4)(np + 6) + \text{tr}_1(\tfrac{1}{2}\Omega)(np + 2)(np + 4) \\
&\cdot (np + 6)(np + 8) + 1/2! \text{tr}_1^2(\tfrac{1}{2}\Omega)(np + 4)(np + 6)(np + 8)(np + 10) \\
&+ 1/3! \text{tr}_1^3(\tfrac{1}{2}\Omega)(np + 6)(np + 8)(np + 10)(np + 12) \dots]
\end{aligned}$$

The fourth moment of $\text{tr } A$ can now be simplified:

$$\begin{aligned}
 E(\text{tr}^4 A) &= \sigma^8 \text{np}(\text{np}+2)(\text{np}+4)(\text{np}+6) \text{etr}(-\frac{1}{2}\Omega) \\
 &\quad \cdot \left\{ (1 + (\text{tr}_1 \frac{1}{2}\Omega) + 1/2! \text{tr}_1^2(\frac{1}{2}\Omega) + \dots) \right. \\
 &\quad + \frac{4}{\text{np}} \text{tr}_1(\Omega) (1 + \text{tr}_1(\frac{1}{2}\Omega) + 1/2! \text{tr}_1^2(\frac{1}{2}\Omega) + 1/3! \text{tr}_1^3(\frac{1}{2}\Omega) + \dots) \\
 &\quad + \frac{6}{\text{np}(\text{np}+2)} \text{tr}_1^2(\Omega) (1 + \text{tr}_1(\frac{1}{2}\Omega) + 1/2! \text{tr}_1^2(\frac{1}{2}\Omega) + 1/3! \text{tr}_1^3(\frac{1}{2}\Omega) + \dots) \\
 &\quad + \frac{4}{\text{np}(\text{np}+2)(\text{np}+4)} \text{tr}_1^3(\Omega) (1 + \text{tr}_1(\frac{1}{2}\Omega) + 1/2! \text{tr}_1^2(\frac{1}{2}\Omega) \\
 &\quad + 1/3! \text{tr}_1^3(\frac{1}{2}\Omega) + \dots) \\
 &\quad \left. + \frac{1}{\text{np}(\text{np}+2)(\text{np}+4)(\text{np}+6)} \text{tr}_1^4(\Omega) (1 + \text{tr}_1(\frac{1}{2}\Omega) \right. \\
 &\quad \left. + 1/2! \text{tr}_1^2(\frac{1}{2}\Omega) + 1/3! \text{tr}_1^3(\frac{1}{2}\Omega) + \dots) \right\} \\
 &= \sigma^8 \{ \text{np}(\text{np}+2)(\text{np}+4)(\text{np}+6) + 4(\text{np}+2)(\text{np}+4)(\text{np}+6) \text{tr}_1(\Omega) \\
 &\quad + 6(\text{np}+4)(\text{np}+6) \text{tr}_1^2(\Omega) + 4(\text{np}+6) \text{tr}_1^3(\Omega) + \text{tr}_1^4(\Omega) \} \\
 (2.9) \quad &= \sigma^8 \sum_{i=0}^4 \binom{4}{i} 2^{4-i} \left(\frac{\text{np}}{2} + i \right)^{(4-i)} \text{tr}_1^i(\Omega)
 \end{aligned}$$

A similar expression for each of the first three moments was derived by using the same method.

Thus a general expression for $E(\text{tr}_1^h A)$ for $h = 1, 2, 3, 4$ is:

$$(2.10) \quad E(\text{tr}_1^h A) = \sigma^{2h} \sum_{i=0}^h \binom{h}{i} 2^{h-i} \left(\frac{\text{np}}{2} + i \right)^{(h-i)} \text{tr}_1^i \Omega.$$

(2.10) can be proved by induction as follows:

We know that:

$$\begin{aligned}
 T^* &= E(\text{tr}^h(A) \text{etr}(\beta A)) \\
 &= E(\text{tr}^h A) + \beta E(\text{tr}^{h+1} A) + (\beta^2/2!) E(\text{tr}^{h+2} A) + \dots
 \end{aligned}$$

Furthermore:

$$\begin{aligned}
 T^* &= (1/\Gamma_p(\frac{1}{2}n)) |2\sigma^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega) / (C_K(I_p)(\frac{n}{2})_K) 2^k \sigma^{2k} k! \\
 &\quad \int_{A>0} |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}\sigma^2 A) \text{etr}(\beta A) \sum_{H \subseteq Q} \sum_{H \subseteq Q} g_{H,K} C_Q(A) dA \\
 &= (1 - 2\beta\sigma^2)^{-\frac{1}{2}np} (1/\Gamma_p(\frac{1}{2}n)) |2\lambda^2 I_p|^{-\frac{1}{2}n} \text{etr}(-\frac{1}{2}\Omega) \\
 &\quad \sum_{k=0}^{\infty} \sum_K C_K(\frac{1}{2}\Omega) / (C_K(I_p)(\frac{n}{2})_K) k! (1 - 2\beta\sigma^2)^k (2\lambda^2)^k.
 \end{aligned}$$

$$\begin{aligned}
& \int_{A>0} |A|^{1/2(n-p-1)} \text{etr}(-1/2 \lambda^{-2} A) \sum_{H, Q} \sum_{K}^Q C_Q(A) dA, \text{ where } \lambda^2 = \sigma^2 / (1 - 2\beta\sigma^2) \\
&= (1 - 2\beta\sigma^2)^{-1/2 np} \text{etr}(-1/2 \Omega) \text{etr}(1/2 \Omega^*) (1/\Gamma_p(1/2 n)) |2\lambda^2 I_p|^{-1/2 n} \text{etr}(-1/2 \Omega^*) \\
&\cdot \sum_{k=0}^{\infty} \sum_K C_K(1/2 \Omega^*) / (C_K(I_p) \binom{n}{2} K(2\lambda^2)^k k!) \int_{A>0} |A|^{1/2(n-p-1)} \text{etr}(-1/2 \lambda^{-2} A) \\
&\quad \sum_{H, Q} \sum_{K}^Q C_Q(A) dA,
\end{aligned}$$

$$\begin{aligned}
& \text{where } \Omega^* = \Omega / (1 - 2\beta\sigma^2) \text{ and } C_K(1/2 \Omega) (1 - 2\beta\sigma^2)^{-k} = C_K(1/2 \Omega / (1 - 2\beta\sigma^2)) \\
&= (1 - 2\beta\sigma^2)^{-1/2 np} \text{etr}\{(-1/2 \Omega) + (1/2 \Omega^*)\} \lambda^{2h} \sum_{i=0}^h \binom{h}{i} 2^{h-i} \left(\frac{np+i}{2}\right)^{(h-i)} \text{tr}_1^i(\Omega^*), \\
&\quad \text{by using 2.3 and 2.10} \\
&= (1 - 2\beta\sigma^2)^{-(1/2 np + h)} \sigma^{2h} \text{etr}(\Omega \beta \sigma^2 / (1 - 2\beta\sigma^2)) \sum_{i=0}^h \binom{h}{i} 2^{h-i} \left(\frac{np+i}{2}\right)^{(h-i)} \text{tr}_1^i(\Omega^*).
\end{aligned}$$

Thus:

$$\begin{aligned}
E(\text{tr}^{h+1} A) &= \left. \frac{d\Gamma^*}{d\beta} \right|_{\beta=0} \\
&= \sigma^{2(h+1)} \left(2 \binom{np}{2} + h \right) + \text{tr}_1(\Omega) \sum_{i=0}^h \binom{h}{i} 2^{h-i} \left(\frac{np+i}{2}\right)^{(h-i)} \text{tr}_1^i(\Omega) \\
&\quad + \sum_{i=0}^h \binom{h}{i} 2^{h+1-i} \left(\frac{np+i}{2}\right)^{(h-i)} \text{tr}_1^i(\Omega) \\
&= \sigma^{2(h+1)} \left(\binom{h}{0} 2^{h+1} \left(\frac{np}{2}\right)^{(h+1)} + \sum_{r=1}^h \text{tr}_1^r(\Omega) 2^{h-r+1} \left\{ \binom{h}{r} \left(\frac{np}{2} + r\right)^{(h-r+1)} \right. \right. \\
&\quad \left. \left. + \binom{h}{r-1} \left(\frac{np}{2} + r - 1\right)^{(h-r+1)} + \text{tr}_1^r(\Omega) \left(\frac{np}{2} + r\right)^{(h-r)} \right\} + \binom{h}{h} \text{tr}_1^{h+1}(\Omega) \right) \\
&= \sigma^{2(h+1)} \left(\binom{h+1}{0} 2^{h+1} \left(\frac{np}{2}\right)^{(h+1)} + \sum_{r=1}^h \text{tr}_1^r(\Omega) 2^{h-r+1} + \binom{h}{r} \left(\frac{np}{2} + r\right)^{(h-r+1)} \right. \\
&\quad \left. + \binom{np}{2} + r\right)^{(h-r)} \left(\frac{np}{2} + h\right) \left\{ \binom{h+1}{h+1} \text{tr}_1^{h+1}(\Omega) \right\} \\
&= \sigma^{2(h+1)} \left(\binom{h+1}{0} 2^{h+1} \left(\frac{np}{2}\right)^{(h+1)} + \sum_{r=1}^h \text{tr}_1^r(\Omega) 2^{h+1-r} \binom{h+1}{r} \left(\frac{np}{2} + r\right)^{(h+1-r)} \right. \\
&\quad \left. + \binom{h+1}{h+1} \text{tr}_1^{h+1}(\Omega) \right) \\
&= \sigma^{2(h+1)} \sum_{i=0}^{h+1} \binom{h+1}{i} 2^{h+1-i} \left(\frac{np}{2} + i\right)^{h+1-i} \text{tr}_1^i(\Omega).
\end{aligned}$$

The following follow as special cases of (2.10):

For $h = 1$ de Waal (1970) proved for general Σ :

$$(2.11) \quad E(\text{tr}_1 A) = n \text{tr}_1(\Sigma) + \text{tr}_1(\Sigma \Omega)$$

Thus when $\Sigma = \sigma^2 I$ we have:

$$\begin{aligned} E(\text{tr}_1 A) &= np \sigma^2 + \sigma^2 \text{tr}_1(\Omega) \\ &= \sigma^2 \left(2 \binom{np}{2} + \text{tr}_1(\Omega) \right) \quad \text{as in (2.10) for } h = 1. \end{aligned}$$

If we consider A with a central Wishart distribution with general covariance matrix Σ , then it follows that: (Khatri and Pillai 1967)

$$(2.12) \quad E(\text{tr}_1^2 A) = n(n+2) \text{tr}_1^2 \Sigma - 4n \text{tr}_2 \Sigma$$

$$(2.13) \quad E(\text{tr}_1^3 A) = n(n+2)(n+4) \text{tr}_1^3 \Sigma - 12n(n+2) \text{tr}_1 \Sigma \text{tr}_2 \Sigma + 24n \text{tr}_3 \Sigma$$

$$\begin{aligned} (2.14) \quad E(\text{tr}_1^4 A) &= n(n+2)(n+4)(n+6) \text{tr}_1^4 \Sigma - 24n(n+2)(n+4) \text{tr}_2 \Sigma \text{tr}_1^2 \Sigma \\ &\quad + 48n(n+2) \text{tr}_2^2 \Sigma + 96n(n+2) \text{tr}_1 \Sigma \text{tr}_3 \Sigma - 192n \text{tr}_4 \Sigma \end{aligned}$$

If $\Sigma = \sigma^2 I$ these formulae reduce respectively to:

$$(2.15) \quad E(\text{tr}_1^2 A) = \sigma^4 np(np+2)$$

$$(2.16) \quad E(\text{tr}_1^3 A) = \sigma^6 np(np+2)(np+4)$$

$$(2.17) \quad E(\text{tr}_1^4 A) = \sigma^8 np(np+2)(np+4)(np+6)$$

which is the same as (2.10) for $h = 2, 3, 4$ and $\Omega = 0$.

Acknowledgement

The author wishes to express his thanks to Prof. D.J. de Waal for valuable suggestions.

REFERENCES

- CONSTANTINE, A.G. (1963). Some non-central distribution problems in multivariate analysis. *Ann. Math. Statist.* **34**, 1270-1285.
- DE WAAL, D.J. (1970). On Bayesian estimation in multivariate analysis. Submitted for publication in *Ann. Math. Statist.*
- HERZ, C.S. (1955). Bessel functions of matrix argument. *Ann. Math.* **61**, 474-523.
- JAMES, A.T. (1964). Distributions of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.* **35**, 475-501.
- KHATRI, C.G. and PILLAI, K.C.S. (1967). On the moments of traces of two matrices in multivariate analysis. *Ann. Inst. Statist. Math.* **19**, 143-156.

KHATRI, C.G. and PILLAI, K.C.S. (1968). On the moments of elementary symmetric functions of the roots of two matrices and approximations to a distribution. *Ann. Math. Statist.* 39, 1274-1281.

Manuscript received 30th September 1970.